SUR4350C Advanced Photogrammetry, SUR6355 Analytical Photogrammetry

Fall, 2025
Hybrid, 3 Credits

Dr. Ben Wilkinson
305 Reed Lab
352 392-3465
benew@ufl.edu

Office Hours: Wednesday (3:00–5:00PM) Thursdays (1:30–2:30PM) in-person or via web conferencing. Please schedule an appointment by emailing the instructor to ensure availability.

Class Meeting Times
Lectures T, R 5th period (11:45-12:35), Labs R 8-10 period (3:00-6:00)

Course Description
Precise photogrammetric measurements, camera calibration, object space coordinate systems, analytical control extension, stereoplotter mapping, digital mapping, and softcopy stereoplotters.

Course Learning Objectives
This course relates the principles of precise measurement and proper data reduction through measurements of photographs followed by calculations to determine spatial information. At the completion of the course, the student should be able to:

- Apply the principles of precise measurement and data reduction through measurements of photographs
- Utilize current methods and standards of photogrammetric practice to solve spatial problems
- Explain the theoretical basis underlying the methods used in solving photogrammetry problems.

Course Overview and Purpose
The course is scaffolded by ten modules that build upon each other as the semester progresses. Concepts from SUR3331C and SUR3520 are combined and applied to fundamental geospatial image processing. Software changes quickly, fundamental concepts do not. With this in mind, and although contemporary software will be used in exercises, the course focuses mostly on theory.

Course Prerequisites
Meas. Science SUR3520, Photogrammetry SUR3331C.

Textbook
Wolf, P.R., B.A. Dewitt, B.E. Wilkinson Elements of Photogrammetry: with Applications in GIS, 4th Ed., McGraw-Hill, 2014.

Interaction Plan
Gainesville section students are required to attend lectures and labs in-person unless accommodation is otherwise discussed and agreed upon with the instructor.

Online section students may attend live lectures and lab sessions via Zoom or may watch the recordings of these sessions when they are posted to the Canvas website. While lab recordings and posted material provide enough guidance to complete the labs, synchronous online attendance is encouraged.

Students may access lecture notes and recordings, assignments, readings, and supporting materials through the Canvas course website as they become available. Students are strongly encouraged to contact the instructor for assistance needed on assignments. Guidance may be provided during office hours, over Canvas messaging, or through ad hoc Zoom sessions.

Synchronous online sessions may be recorded. By sharing your video, screen, or audio during any synchronous online class sessions, you are consenting to being recorded for the benefit of students who cannot attend live as well as for class review during the current semester. If you have special circumstances or concerns about privacy, it is your responsibility to discuss it with your instructor.

Students should expect posted grades and feedback on assignments within approximately one week from the due date.

Required Technology

- A computer or mobile device with high-speed internet connection.
- A webcam, headset and/or microphone, and speakers.
- Latest version of web browser. Canvas supports only the two most recent versions of any given browser. What browser am I using?
- Installation of proctoring software may be required and will be provided if so.

Communication Guidelines

Online communication with the instructor should be through the Canvas website via the course discussion boards or Canvas email (Inbox).

Class Demeanor/Expectations

Just as in any professional environment, meaningful and constructive dialogue is expected in this class and requires a degree of mutual respect, willingness to listen, and tolerance of opposing points of view. Respect for individual differences and alternative viewpoints will be maintained in this class at all times. All members of the class are expected to follow rules of common courtesy, decency, and civility in all interactions. Failure to do so will not be tolerated and may result in loss of participation points and/or referral to the Dean of Students' Office.

Technical Support

UF Computing Help Desk & Ticket Number: All technical issues require a UF Helpdesk Ticket Number. The UF Helpdesk is available 24 hours a day, 7 days a week. https://helpdesk.ufl.edu/ | 352-392-4357

Modules

Mod	# Lec	Topics	Reading	Reading Quiz
1	1	**Introduction**	**Syllabus**	
2	1	**3D angles and angular conversion**	**C-7 (Step 1 only), D-9, D-11**	**Quiz 1**
3	3	**2D Coordinate transformations**	**C-1…C-6, C9, C10**	**Quiz 2**
4	2	**3D Conformal coordinate transformation**	**Section C-7 (all steps), C-8**	**Quiz 3**
5	1	**Homogeneous coordinate representation**	**C-11**	**Quiz 4**
6	5	**Collinearity**, space resection, space intersection, relative orientation, analytical stereomodel	**Ch. 11**	**Quiz 5** **Quiz 6**
7	5	**Aerotriangulation** - Independent Models, bundle adjustment, self-calibration, airborne control	**Ch. 17, 3-9, 3-10, 19-4…19-9**	**Quiz 7** **Quiz 8**
8	4	**Digital photogrammetry** - Modulation transfer, image model, spatial frequency, resampling, softcopy, image matching, orthophoto generation, digital mapping cameras, ASPRS accuracy standards	**3-14, Ch. 15, App. E**	**Quiz 9** **Quiz 10**
9	4	**Lidar**	**Ch. 14**	**Quiz 11**
10	1	**Computer Vision,** UAS Photogrammetry, Synthesized Concepts		

Weekly Course Schedule

Week	Topic	Assessment	Due Dates
1	Module 1 Introduction	Discussion Board Introductions	Aug. 25
2	Module 2, 3	Lab 1 (3D Rotation) Quiz 1	Sep. 3 Aug. 29
3	Module 3	Lab 2 (2D Transformations) Quiz 2	Sep. 10 Sep. 5
4	Module 4	Lab 3 (3D transformations, Lidar) Quiz 3	Sep. 24 Sep. 12
5	Module 5, 6	Quiz 4	Sep. 19
6	Module 6	Lab 4 (Space Resection) Quiz 5	Oct. 1 Sep. 26
7	Module 6	Quiz 6 **Exam 1**	Oct. 1 **Oct. 2**
8	Module 7	Lab 5 (Bundle Adjustment) Quiz 7	Oct. 29 Oct. 10
9	Module 7	Quiz 8	Oct. 18
10	Module 7, 8	Lab 6 (Stereoplotter) Quiz 9	Nov. 12 Oct. 24
11	Module 8	Quiz 10	Oct. 31

Week	Topic	Assessment	Due Dates
12	Module 8	Lab 7 (Self Calibration) **Exam 2**	Nov. 20 Nov. 6
13	Module 9	Quiz 11	Nov. 21
14	Module 9, 10		
15	Module 10		
		FINAL EXAM	Dec. 9 (3-5 PM)

Grading Policy

Course grading is consistent with UF grading policies.

Course Grading Structure

The course grade will be based on participation, labs, reading assignment quizzes, computer programming homework (SUR 6355) and three exams. The percentage breakdown is as follows (subject to change based on course progression). Final grades are based on a curve.

Assignment Type	Point Value	Percent of Final Grade
Participation	5	5%
Labs	10-20 points	35%
Quizzes	5 points	20%
Exam 1	100	12.5%
Exam 2	100	12.5%
Final Exam	100	15%

Grading Scale

Grade	Percentage
A	94-100
A-	90-93.99
B+	87-89.99
B	83-86.99
B-	80-82.99
C+	77-79.99
C	73-76.99
C-	70-72.99
D+	67-69.99

Grade	Percentage
D	63-66.99
D-	60-62.99
E	0-59.99

Late Policy

Due dates will be announced when assignments are given. A deduction of 10% will be made for each day they are late. It is the responsibility of the student to access on-line lectures, readings, quizzes, and exams and to maintain satisfactory progress in the course. Requirements for class attendance and make-up exams, assignments and other work are consistent with university policies that can be found at: https://catalog.ufl.edu/ugrad/current/regulations/info/attendance.aspx

Policy on AI Use

Although it can be useful as a tool for some things, generative AI can be unreliable and doesn't always provide correct information. On certain assignments, the instructor will explicitly either allow or disallow the use of generative AI, such as ChatGPT, in completing the work. If it is explicitly allowed, students must thoroughly cite any information or assistance received from AI/NLP platforms, just as you must for any resource. If the instructor has not mentioned the use of this technology for a given assignment, the student should assume it is entirely prohibited. When in doubt, students should ask the instructor if and how AI/NLP may be used to avoid the risk of losing up to all points on the associated assignment. This policy is not a substitute for the policy of any other instructor or course. All courses will have their own individual policy, and students should learn and understand the policy for any course they are taking.

Academic Policies and Resources

Academic policies for this course are consistent with university policies. See https://syllabus.ufl.edu/syllabus-policy/uf-syllabus-policy-links/

Campus Health and Wellness Resources

Visit https://one.uf.edu/whole-gator/topics for resources that are designed to help you thrive physically, mentally, and emotionally at UF.

Please contact UMatterWeCare for additional and immediate support.

Software Use

All faculty, staff and students of the university are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against university policies and rules, disciplinary action will be taken as appropriate.

Privacy and Accessibility Policies

[required for online courses, list all technology used]

- Instructure (Canvas)
  - Instructure Privacy Policy
  - Instructure Accessibility

- Zoom
    - Zoom Privacy Policy
    - Zoom Accessibility